Generalized S-X2 compared to conditional infit and item-restscore

Detection of 1-3 misfitting items under multidimensionality

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Purpose

Johansson (2025) compared conditional item infit (with simulation-based cutoffs) and item-restscore for detecting item misfit caused by multidimensionality, and found that neither method behaves well at large sample sizes: false positive rates rise sharply from n = 1000 onwards. Kang and Chen (2008) report that the generalized S-X2 index maintains nominal Type I error rates from n = 500 up to n = 5000, which raises the question of whether it could fill that gap.

Two things make this worth testing rather than assuming:

1. Kang and Chen only evaluated power in conditions where **every** item was misfitting, and where misfit was a clean nested-model contrast (RSM data calibrated as PCM, etc.). The realistic case of a few misfitting items among fitting ones was explicitly left for future work.
2. The follow-up study on the graded response model found S-X2 power to be "much lower when the source of misfit was multidimensionality" than when it came from a wrong item response function. Multidimensionality is precisely the misfit source used in Johansson (2025), and the dominant concern in applied Rasch analysis.

This simulation therefore replicates the Johansson (2025) design and adds S-X2 as a fourth method, so all methods are evaluated on the *same* datasets.

Design

Following Johansson (2025):

- 20 dichotomous items, locations drawn from `runif(20, -2, 2)`
- person locations from a bivariate normal, SD 1.5, correlation 0.15 between dimensions (lower correlation = stronger multidimensionality)
- misfit induced with `eRm::sim.xdim()` by loading selected items on the second dimension
- misfitting items chosen at approximately 0, −1 and −2 logits relative to the sample mean, to vary targeting
- master datasets of 10 000 respondents, from which samples are drawn
- sample sizes 150, 250, 500, 1000, 2000

Three misfit conditions are used: **0** misfitting items (a clean null, to measure Type I error uncontaminated by other misfitting items), **1** (the on-target item only), and **3**.

Four decision rules are compared:

Label	Rule
`infit`	conditional infit MSQ outside simulation-based cutoffs (`RMitemInfitCutoff`, 100 iterations for n < 500, 200 otherwise)
`restscore`	item-restscore gamma, BH-corrected p < .05
`sx2_bh`	generalized S-X2, BH-corrected p < .05
`sx2_rmsea`	RMSEA.S_X2 > .05 (effect size, no significance test)

Uncorrected S-X2 p-values are also recorded, since infit cutoffs are not multiplicity-corrected and the comparison should be readable either way.

S-X2 is computed via `mirt::itemfit()`, which implements the Kang and Chen generalization. Note that this estimates item parameters by MML under a normal θ prior, whereas the other two methods use CML — see the discussion at the end.

```
suppressMessages({
  library(mirt)
  library(easyRasch2)
  library(eRm)
  library(MASS)
  library(parallel)
  library(dplyr)
  library(tidyr)
  library(ggplot2)
})

R_REPS   <- 200
NS       <- c(150, 250, 500, 1000, 2000)
NMISFITS <- c(0, 1, 3)
NCORES   <- 12   # performance cores only (M4 Max)
RES_FILE <- "sx2_itemfit_sim_results.rds"
```

Data generation

```
make_master <- function(n_misfit, N = 10000, seed = 20250727) {
  set.seed(seed)
  locs <- runif(20, -2, 2)
  # items closest to 0, -1, -2 logits (sample theta mean = 0)
  misfit_idx <- integer(0)
  for (tg in c(0, -1, -2)) {
    cand <- setdiff(order(abs(locs - tg)), misfit_idx)[1]
    misfit_idx <- c(misfit_idx, cand)
  }
  use <- misfit_idx[seq_len(n_misfit)]
  W <- cbind(rep(1, 20), rep(0, 20))
  if (n_misfit > 0) W[use, ] <- matrix(c(0, 1), nrow = length(use), ncol = 2,
                                       byrow = TRUE)
  Sigma <- matrix(c(1.5^2, .15 * 1.5^2, .15 * 1.5^2, 1.5^2), 2, 2)
  set.seed(seed + n_misfit)
  th <- MASS::mvrnorm(N, mu = c(0, 0), Sigma = Sigma)
  d <- as.data.frame(eRm::sim.xdim(persons = th, items = locs,
                                   Sigma = Sigma, weightmat = W))
  names(d) <- paste0("V", 1:20)
  list(data = d, locs = locs, misfit = use, misfit_all = misfit_idx)
}

masters <- lapply(NMISFITS, make_master)
names(masters) <- as.character(NMISFITS)
misfit_all <- masters[["3"]]$misfit
locs <- masters[["3"]]$locs

data.frame(
  item = paste0("V", misfit_all),
```

```
  intended_location = c(0, -1, -2),
  actual_location = round(locs[misfit_all], 2)
)
```

item	intended_location	actual_location
V8	0	0.04
V12	−1	−1.00
V11	−2	−2.00

One replication

```
one_rep <- function(rep, k, n) {
  set.seed(1e6 * k + 1e3 * n + rep)
  md <- masters[[as.character(k)]]$data
  d  <- md[sample(nrow(md), n), ]
  iters <- if (n < 500) 100 else 200
  out <- matrix(NA_integer_, nrow = 20, ncol = 5,
                dimnames = list(NULL, c("infit", "restscore", "sx2_raw",
                                        "sx2_bh", "sx2_rmsea")))
  try({
    co <- RMitemInfitCutoff(d, iterations = iters, parallel = FALSE,
                            seed = rep, verbose = FALSE)
    fi <- RMitemInfit(d, cutoff = co, output = "dataframe")
    out[, "infit"] <- as.integer(fi$Flagged[match(names(d), fi$Item)] != "")
  }, silent = TRUE)
  try({
    rs <- RMitemRestscore(d, output = "dataframe", p_adj = "BH")
    out[, "restscore"] <- as.integer(rs$Flagged[match(names(d), rs$Item)] != "")
  }, silent = TRUE)
  try({
    mm <- mirt(d, 1, itemtype = "Rasch", verbose = FALSE)
    f  <- mirt::itemfit(mm, fit_stats = "S_X2")
    i  <- match(names(d), f$item)
    out[, "sx2_raw"]   <- as.integer(f$p.S_X2[i] < .05)
    out[, "sx2_bh"]    <- as.integer(p.adjust(f$p.S_X2, "BH")[i] < .05)
    out[, "sx2_rmsea"] <- as.integer(f$RMSEA.S_X2[i] > .05)
  }, silent = TRUE)
  out
}
```

Run

Results are cached; delete sx2_itemfit_sim_results.rds to force a re-run.

```
grid <- expand.grid(rep = 1:R_REPS, n = NS, k = NMISFITS)

if (file.exists(RES_FILE)) {
  sim <- readRDS(RES_FILE)
} else {
  t0 <- Sys.time()
  res <- mclapply(seq_len(nrow(grid)),
                  function(i) one_rep(grid$rep[i], grid$k[i], grid$n[i]),
                  mc.cores = NCORES)
  sim <- list(res = res, grid = grid, misfit = misfit_all, locs = locs,
              elapsed_min = as.numeric(Sys.time() - t0, "mins"))
  saveRDS(sim, RES_FILE)
}

long <- do.call(rbind, lapply(seq_len(nrow(sim$grid)), function(i) {
  m <- sim$res[[i]]
  data.frame(k = sim$grid$k[i], n = sim$grid$n[i], rep = sim$grid$rep[i],
```

```r
             item = 1:20, m)
})) |>
  pivot_longer(c(infit, restscore, sx2_raw, sx2_bh, sx2_rmsea),
               names_to = "method", values_to = "flag") |>
  mutate(is_misfit = mapply(function(k, it) it %in% sim$misfit[seq_len(k)],
                            k, item))

cat("failed method-replications:", sum(is.na(long$flag)), "of", nrow(long), "\n")
```

```
failed method-replications: 0 of 300000
```

Results

Type I error under a true Rasch model (k = 0)

With no misfitting items present, the flag rate on any item is a false positive.

```r
long |>
  filter(k == 0) |>
  group_by(method, n) |>
  summarise(fp = 100 * mean(flag, na.rm = TRUE), .groups = "drop") |>
  pivot_wider(names_from = n, values_from = fp, names_prefix = "n=") |>
  mutate(across(where(is.numeric), \(x) round(x, 1)))
```

method	n=150	n=250	n=500	n=1000	n=2000
infit	1.6	1.6	1.0	1.0	0.9
restscore	0.7	0.3	0.3	0.1	0.1
sx2_bh	0.7	0.5	0.8	0.7	1.1
sx2_raw	6.9	7.1	8.7	7.1	8.8
sx2_rmsea	26.2	11.1	1.4	0.0	0.0

Detection rate, 1 misfitting item

```r
long |>
  filter(k == 1, is_misfit) |>
  group_by(method, n) |>
  summarise(det = 100 * mean(flag, na.rm = TRUE), .groups = "drop") |>
  pivot_wider(names_from = n, values_from = det, names_prefix = "n=") |>
  mutate(across(where(is.numeric), \(x) round(x, 1)))
```

method	n=150	n=250	n=500	n=1000	n=2000
infit	100.0	100	100	100	100
restscore	97.5	100	100	100	100
sx2_bh	98.5	100	100	100	100
sx2_raw	100.0	100	100	100	100
sx2_rmsea	100.0	100	100	100	100

False positives on the 19 fitting items in the same datasets:

```r
long |>
  filter(k == 1, !is_misfit) |>
  group_by(method, n) |>
  summarise(fp = 100 * mean(flag, na.rm = TRUE), .groups = "drop") |>
  pivot_wider(names_from = n, values_from = fp, names_prefix = "n=") |>
  mutate(across(where(is.numeric), \(x) round(x, 1)))
```

method	n=150	n=250	n=500	n=1000	n=2000
infit	2.5	3.1	2.5	6.5	13.7
restscore	2.5	2.6	2.9	5.6	15.1
sx2_bh	0.4	0.7	0.3	0.5	1.4
sx2_raw	5.4	5.1	4.8	6.0	10.2
sx2_rmsea	22.0	8.8	0.2	0.0	0.0

Detection rate, 3 misfitting items

Broken out by item, since targeting matters.

```
long |>
  filter(k == 3, is_misfit) |>
  mutate(target = c("0 logits", "-1 logits", "-2 logits")[match(item, sim$misfit)]) |>
  group_by(method, target, n) |>
  summarise(det = 100 * mean(flag, na.rm = TRUE), .groups = "drop") |>
  pivot_wider(names_from = n, values_from = det, names_prefix = "n=") |>
  mutate(across(where(is.numeric), \(x) round(x, 1))) |>
  arrange(factor(target, levels = c("0 logits", "-1 logits", "-2 logits")), method)
```

method	target	n=150	n=250	n=500	n=1000	n=2000
infit	0 logits	97.0	100.0	100.0	100	100
restscore	0 logits	87.0	100.0	100.0	100	100
sx2_bh	0 logits	88.0	99.0	100.0	100	100
sx2_raw	0 logits	96.5	99.5	100.0	100	100
sx2_rmsea	0 logits	98.5	99.5	100.0	100	100
infit	−1 logits	89.0	98.5	100.0	100	100
restscore	−1 logits	79.0	98.5	100.0	100	100
sx2_bh	−1 logits	80.5	98.0	100.0	100	100
sx2_raw	−1 logits	94.5	100.0	100.0	100	100
sx2_rmsea	−1 logits	99.0	100.0	100.0	100	100
infit	−2 logits	57.5	81.5	99.5	100	100
restscore	−2 logits	52.0	82.5	99.5	100	100
sx2_bh	−2 logits	54.0	78.5	99.0	100	100
sx2_raw	−2 logits	75.0	87.5	99.5	100	100
sx2_rmsea	−2 logits	89.5	91.5	98.0	100	100

False positives on the 17 fitting items:

```
long |>
  filter(k == 3, !is_misfit) |>
  group_by(method, n) |>
  summarise(fp = 100 * mean(flag, na.rm = TRUE), .groups = "drop") |>
  pivot_wider(names_from = n, values_from = fp, names_prefix = "n=") |>
  mutate(across(where(is.numeric), \(x) round(x, 1)))
```

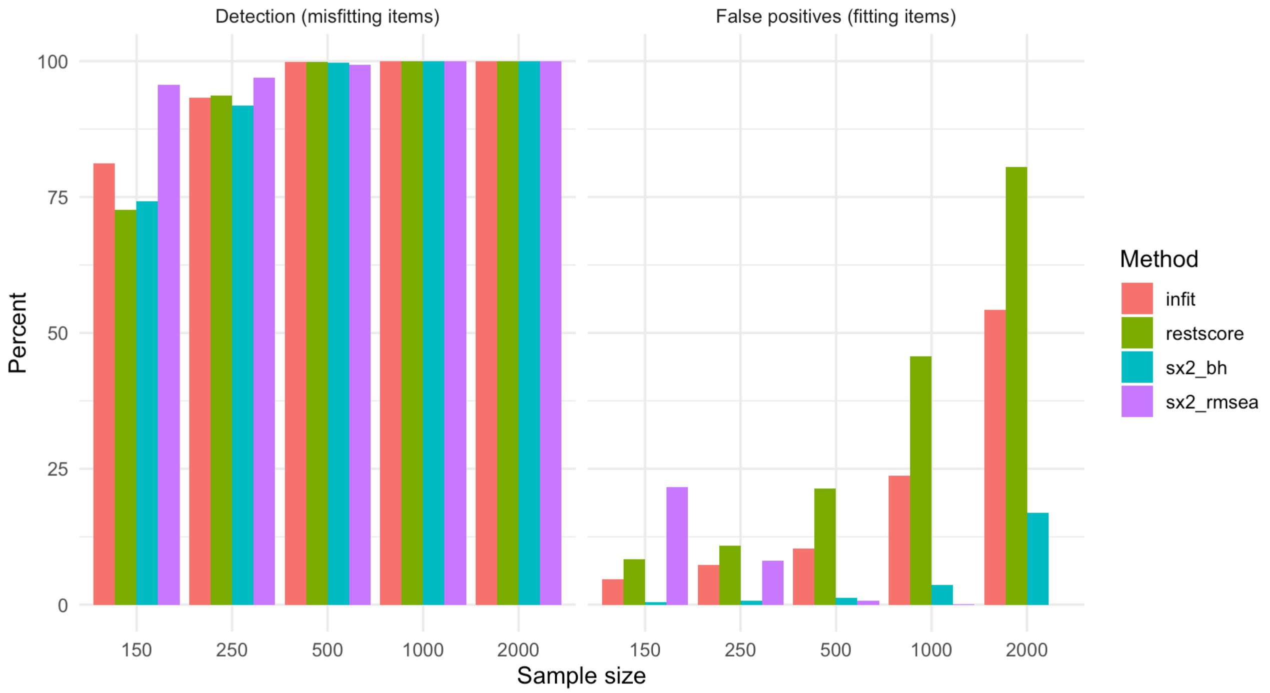
method	n=150	n=250	n=500	n=1000	n=2000
infit	4.7	7.3	10.3	23.7	54.2
restscore	8.4	10.8	21.4	45.7	80.6

method	n=150	n=250	n=500	n=1000	n=2000
sx2_bh	0.5	0.7	1.2	3.6	16.9
sx2_raw	4.8	4.4	6.4	12.6	31.7
sx2_rmsea	21.6	8.1	0.7	0.0	0.0

Figure: detection vs false positives, 3 misfitting items

```
plot_dat <- long |>
  filter(k == 3, method %in% c("infit", "restscore", "sx2_bh", "sx2_rmsea")) |>
  mutate(type = ifelse(is_misfit, "Detection (misfitting items)",
                       "False positives (fitting items)")) |>
  group_by(method, n, type) |>
  summarise(pct = 100 * mean(flag, na.rm = TRUE), .groups = "drop")

ggplot(plot_dat, aes(factor(n), pct, fill = method)) +
  geom_col(position = "dodge") +
  facet_wrap(~type) +
  labs(x = "Sample size", y = "Percent", fill = "Method") +
  theme_minimal(base_size = 12)
```



Supplementary study: weaker multidimensionality

The main study uses an inter-dimensional correlation of 0.15, which is strong misfit — detection saturates at 100% for all methods from n = 500 onwards, so the main study discriminates between methods almost entirely on false positives.

The `sx2_rmsea` rule performs strikingly well at n >= 500 there, but it applies a **fixed** threshold of .05 to an effect size. Its performance therefore depends on how far the true misfit sits above that threshold, and a strong misfit condition flatters it. This supplement re-runs the k = 3 condition with inter-dimensional correlations of 0.50 and 0.75 (progressively weaker multidimensionality) and records the raw `RMSEA.S_X2` values, to check whether the rule survives when the effect is subtle.

```
make_master_r <- function(rho, n_misfit = 3, N = 10000, seed = 20250727) {
  set.seed(seed)
  locs <- runif(20, -2, 2)
  misfit_idx <- integer(0)
  for (tg in c(0, -1, -2)) {
    cand <- setdiff(order(abs(locs - tg)), misfit_idx)[1]
    misfit_idx <- c(misfit_idx, cand)
```

```
  }
  use <- misfit_idx[seq_len(n_misfit)]
  W <- cbind(rep(1, 20), rep(0, 20))
  W[use, ] <- matrix(c(0, 1), nrow = length(use), ncol = 2, byrow = TRUE)
  Sigma <- matrix(c(1.5^2, rho * 1.5^2, rho * 1.5^2, 1.5^2), 2, 2)
  set.seed(seed + round(rho * 100))
  th <- MASS::mvrnorm(N, mu = c(0, 0), Sigma = Sigma)
  d <- as.data.frame(eRm::sim.xdim(persons = th, items = locs,
                                   Sigma = Sigma, weightmat = W))
  names(d) <- paste0("V", 1:20)
  list(data = d, locs = locs, misfit = use)
}

RHOS  <- c(0.50, 0.75)
NS2   <- c(250, 500, 1000, 2000)
R2    <- 150
RES2  <- "sx2_itemfit_sim_weak_results.rds"

masters2 <- lapply(RHOS, make_master_r)
names(masters2) <- as.character(RHOS)

one_rep2 <- function(rep, rho, n) {
  set.seed(1e6 * round(rho * 100) + 1e3 * n + rep)
  md <- masters2[[as.character(rho)]]$data
  d  <- md[sample(nrow(md), n), ]
  iters <- if (n < 500) 100 else 200
  out <- data.frame(item = 1:20, infit = NA_integer_, restscore = NA_integer_,
                    sx2_bh = NA_integer_, sx2_rmsea = NA_integer_,
                    rmsea = NA_real_)
  try({
    co <- RMitemInfitCutoff(d, iterations = iters, parallel = FALSE,
                            seed = rep, verbose = FALSE)
    fi <- RMitemInfit(d, cutoff = co, output = "dataframe")
    out$infit <- as.integer(fi$Flagged[match(names(d), fi$Item)] != "")
  }, silent = TRUE)
  try({
    rs <- RMitemRestscore(d, output = "dataframe", p_adj = "BH")
    out$restscore <- as.integer(rs$Flagged[match(names(d), rs$Item)] != "")
  }, silent = TRUE)
  try({
    mm <- mirt(d, 1, itemtype = "Rasch", verbose = FALSE)
    f  <- mirt::itemfit(mm, fit_stats = "S_X2")
    i  <- match(names(d), f$item)
    out$sx2_bh    <- as.integer(p.adjust(f$p.S_X2, "BH")[i] < .05)
    out$rmsea     <- f$RMSEA.S_X2[i]
    out$sx2_rmsea <- as.integer(f$RMSEA.S_X2[i] > .05)
  }, silent = TRUE)
  out
}

grid2 <- expand.grid(rep = 1:R2, n = NS2, rho = RHOS)

if (file.exists(RES2)) {
  sim2 <- readRDS(RES2)
} else {
  res2 <- mclapply(seq_len(nrow(grid2)),
                   function(i) one_rep2(grid2$rep[i], grid2$rho[i], grid2$n[i]),
                   mc.cores = NCORES)
  sim2 <- list(res = res2, grid = grid2, misfit = masters2[["0.5"]]$misfit)
  saveRDS(sim2, RES2)
}

long2 <- do.call(rbind, lapply(seq_len(nrow(sim2$grid)), function(i)
```

```r
  cbind(sim2$grid[i, ], sim2$res[[i]]))) |>
  mutate(is_misfit = item %in% sim2$misfit)
```

Detection of the three misfitting items:

```r
long2 |>
  filter(is_misfit) |>
  pivot_longer(c(infit, restscore, sx2_bh, sx2_rmsea),
               names_to = "method", values_to = "flag") |>
  group_by(rho, method, n) |>
  summarise(v = 100 * mean(flag, na.rm = TRUE), .groups = "drop") |>
  pivot_wider(names_from = n, values_from = v, names_prefix = "n=") |>
  mutate(across(where(is.numeric), \(x) round(x, 1))) |>
  arrange(rho, method)
```

rho	method	n=250	n=500	n=1000	n=2000
0.5	infit	63.6	83.8	96.4	100.0
0.5	restscore	42.9	81.8	98.2	100.0
0.5	sx2_bh	50.2	77.3	96.9	100.0
0.5	sx2_rmsea	78.9	77.1	73.8	70.7
0.8	infit	23.8	36.4	69.6	94.7
0.8	restscore	2.9	16.7	57.3	93.8
0.8	sx2_bh	9.3	23.3	39.6	77.8
0.8	sx2_rmsea	40.4	24.9	6.9	0.4

False positives on the 17 fitting items:

```r
long2 |>
  filter(!is_misfit) |>
  pivot_longer(c(infit, restscore, sx2_bh, sx2_rmsea),
               names_to = "method", values_to = "flag") |>
  group_by(rho, method, n) |>
  summarise(v = 100 * mean(flag, na.rm = TRUE), .groups = "drop") |>
  pivot_wider(names_from = n, values_from = v, names_prefix = "n=") |>
  mutate(across(where(is.numeric), \(x) round(x, 1))) |>
  arrange(rho, method)
```

rho	method	n=250	n=500	n=1000	n=2000
0.5	infit	3.0	3.4	6.7	15.4
0.5	restscore	3.4	5.9	12.0	25.5
0.5	sx2_bh	0.5	0.9	1.3	2.3
0.5	sx2_rmsea	8.5	0.5	0.0	0.0
0.8	infit	2.5	1.5	2.2	4.4
0.8	restscore	1.2	1.4	2.5	4.4
0.8	sx2_bh	0.7	0.9	0.9	1.8
0.8	sx2_rmsea	9.7	0.9	0.0	0.0

Median `RMSEA.S_X2` for misfitting vs fitting items, against the .05 threshold:

```r
long2 |>
  group_by(rho, n, type = ifelse(is_misfit, "misfitting", "fitting")) |>
  summarise(median_rmsea = round(median(rmsea, na.rm = TRUE), 3),
```

```
            .groups = "drop") |>
  pivot_wider(names_from = n, values_from = median_rmsea, names_prefix = "n=")
```

rho	type	n=250	n=500	n=1000	n=2000
0.50	fitting	0.006	0.004	0.007	0.007
0.50	misfitting	0.075	0.065	0.061	0.058
0.75	fitting	0.000	0.005	0.001	0.003
0.75	misfitting	0.043	0.037	0.032	0.030

Conclusions

1. S-X2 has no power advantage, and a power disadvantage against subtle misfit. With strong multidimensionality (rho = 0.15) all methods are equivalent and saturate. With rho = 0.75, conditional infit clearly dominates: 23.8 / 36.4 / 69.6 / 94.7 percent detection at n = 250 / 500 / 1000 / 2000, against 9.3 / 23.3 / 39.6 / 77.8 for S-X2. The concern raised by the Kang and Chen GRM follow-up — weak power against multidimensionality — does not appear under strong misfit but is clearly visible once the misfit is subtle.

2. S-X2 does control the false positive cascade. This is its real contribution. With three strongly misfitting items at n = 2000, false positives on fitting items are 54.2 percent for infit and 80.6 percent for item-restscore, against 16.9 percent for S-X2. On the practically relevant metric — recovering all three misfitting items with no false positives — S-X2 achieves 72 percent at n = 1000-2000 with rho = 0.5, where infit manages 26 and 3.3 percent.

3. Infit and item-restscore are not miscalibrated. Under a true Rasch model with no misfitting items, false positive rates are 0.1-1.6 percent at every sample size, including n = 2000. The large-sample false positives reported in Johansson (2025) are therefore entirely the overfit cascade caused by the misfitting items themselves, not a defect in the simulation-based cutoffs.

4. A fixed RMSEA threshold of .05 is not usable. It fails in both directions. At n = 150 under a true model it flags 26.2 percent of items, because the null RMSEA has not yet shrunk. At rho = 0.75 the median RMSEA of genuinely misfitting items is .030-.043 — below the threshold — so detection falls to 0.4 percent at n = 2000, and gets *worse* as n grows, since the estimate concentrates around a true value under the cutoff.

5. The implied design. RMSEA.S_X2 has the property that is wanted — it does not inflate with sample size (median .058-.075 for misfitting items across n = 250-2000 at rho = 0.5) — but the threshold is wrong and is itself sample-size dependent. Median RMSEA for fitting items is .000-.007 at n >= 250, far below .05. This is exactly the problem `RMitemInfitCutoff` solves for infit, which suggests a simulation-based cutoff for RMSEA.S_X2 would set a much lower and n-appropriate threshold, and would likely recover the rho = 0.75 detection that the fixed rule misses. **This is a hypothesis, not a result — it has not been tested here.**

Overall: S-X2 is a large-sample complement to conditional infit, not a replacement for it. Its niche is several strongly misfitting items at n >= 1000, which is precisely where the existing methods fail.

Notes on interpretation

`sx2_rmsea` uses a fixed RMSEA cutoff of .05, borrowed from SEM convention. Item-level RMSEA cutoffs are not well established, and the null RMSEA shrinks with sample size, so this rule is expected to behave asymmetrically across n. It is included to separate two questions that the S-X2 significance test confounds: whether the departure is detectable, and whether it is large.

A caveat on estimation: `mirt::itemfit()` estimates item parameters by MML under a normal θ prior. For the Rasch family the S-X2 expected proportions are in principle distribution-free — the θ density cancels between numerator and denominator, leaving the CML conditional probabilities — but that only holds at the true parameters. A CML-based implementation would remove the normality assumption, which matters here since the generating θ distribution is normal and therefore favourable to the mirt route.

References

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